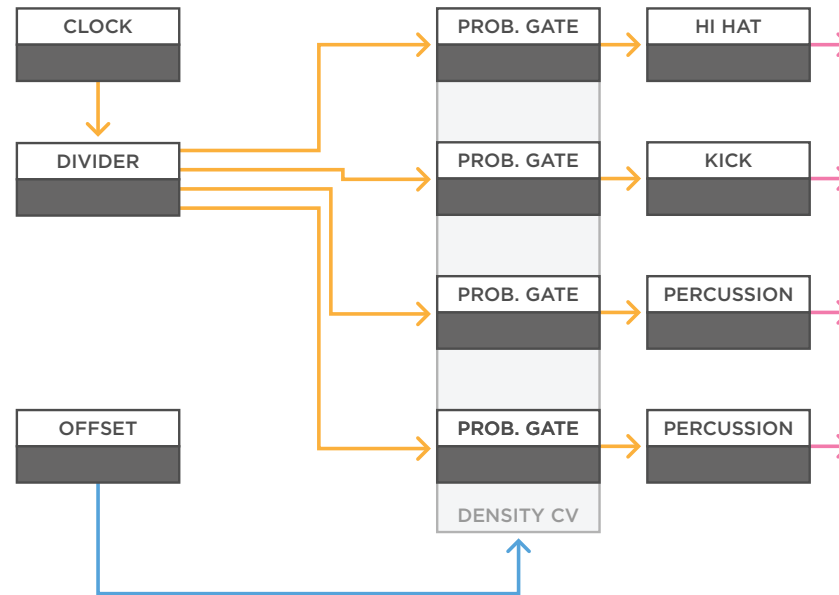


Probabilistic Gates - CuteLab MissedOps

Trigger Patch

MONOTRAIL TECH TALK

- Audio
- Control Voltage
- Trigger / Gate



(a)

a - In a modular system triggers and gates can be used on pretty much anything you like. Therefore adding randomization to patterns with a probabilistic gate, can be useful in many situations. To make a simple start, you can take a steady clock, feed it into a clock divider, and send different divisions to each of the probabilistic gates.

For example, a 16th step division to some hats, a quarter division to a kick, and two other divisions to percussive sounds. Now, if you feed the density CV a manual offset voltage you create hands on control over the pattern.

MONOTRAIL TECH TALK

```

graph LR
    Seq[SEQUENCER  
Trigger] -- Orange --> PG1[PROB. GATE]
    Seq -- Orange --> PG2[PROB. GATE]
    Seq -- Orange --> PG3[PROB. GATE]
    Seq -- Orange --> PG4[PROB. GATE]
    
    Env[ENVELOPE  
✓] -- Blue --> Mix[MIXER]
    
    Off1[OFFSET] -- Blue --> Mix
    Off2[OFFSET] -- Blue --> Mix
    
    Mix -- Blue --> DCV[DENSITY CV]
    
    PG1 -- Orange --> HH[HI HAT]
    PG2 -- Orange --> K[KICK]
    PG3 -- Orange --> P1[PERCUSSION]
    PG4 -- Orange --> P2[PERCUSSION]
    
    HH -- Pink --> Out1[...]
    K -- Pink --> Out2[...]
    P1 -- Pink --> Out3[...]
    P2 -- Pink --> Out4[...]

```

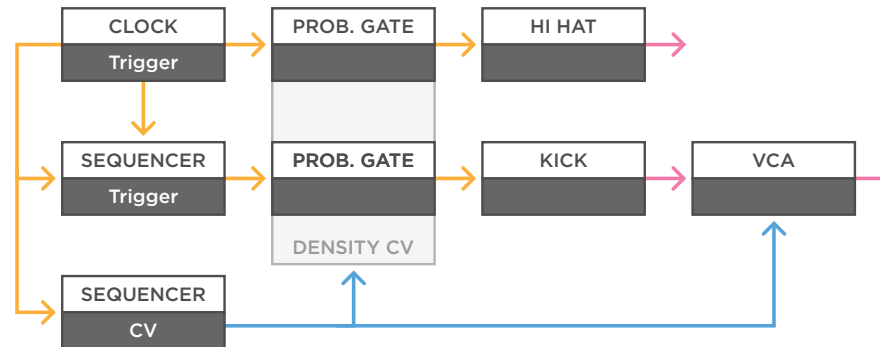
a - But you can use trigger sequences as well. For example, you can use multiple sequences from a beatstep Pro to create more interesting patterns. To create more controlled temporary random variations, you can modulate the density CV over time. For example, you can use a another trigger, perhaps one that occurs only once every two bars, and have it trigger an envelope.

Monotrail Tech Talk #33 - How to spice up patterns with probabilistic gates 2/8

Probabilistic Gates - CuteLab MissedOps

Trigger Patch

- Audio
- Control Voltage
- Trigger / Gate



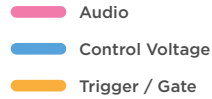
(a)

a - You can create step based probability in combination with a CV sequencer. For example, use a clock to progress a simple 8 step trigger sequencer, and program a pattern triggering the kick on the 1st, 5th and 8th step. Then, also send the clock output to trigger a hat on every step. Now when you use the same clock on an 8 step CV sequencer, and send that sequence to the Density input, now you can dial in the probability for each step.

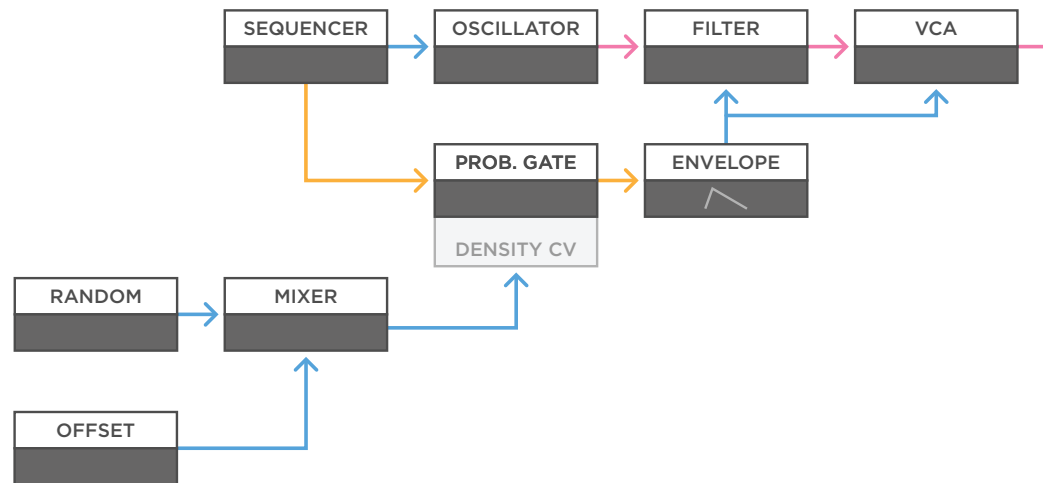
For example, set the probability on the 1st and 5th step to be 100 percent, so the core beat stays up, but set other probabilities for the other steps. You can send the hats through a VCA, and modulate that with the cv sequence as well to add some volume modulation.

Probabilistic Gates - CuteLab MissedOps

Melodic Patch



MONOTRAIL TECH TALK



(a)

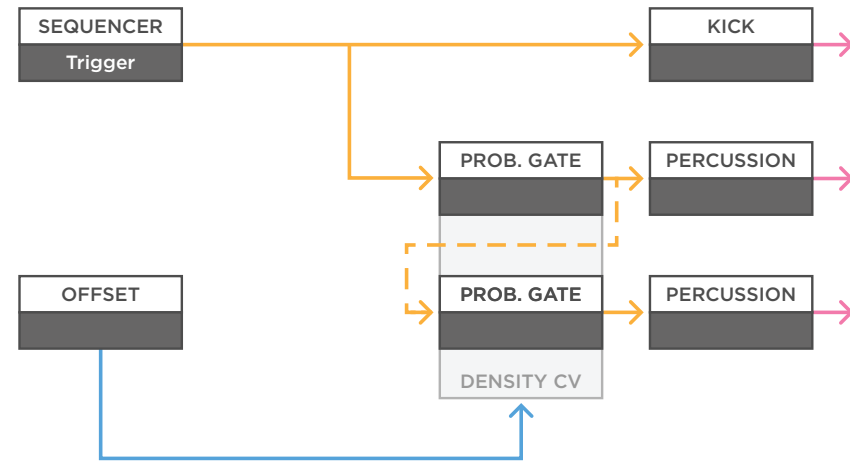
a - All the previous tricks can be used on other things than percussive sounds. For example to spice up a simple melodic voice, made with an oscillator, filter and VCA. When you use an 8 steps sequencer to create a looping melody, you can send the gate output to a probabilistic gate. Then use the randomized triggers on an envelope, and have that open the filter and VCA.

You can use some modulation to automate the randomization amount. A slow smooth random voltage can really work well here. Again use an offset voltage to dial in the base density, and mix in the random voltage to determine the spread.

Probabilistic Gates - CuteLab MissedOps

Normalled Patches - Match / Miss

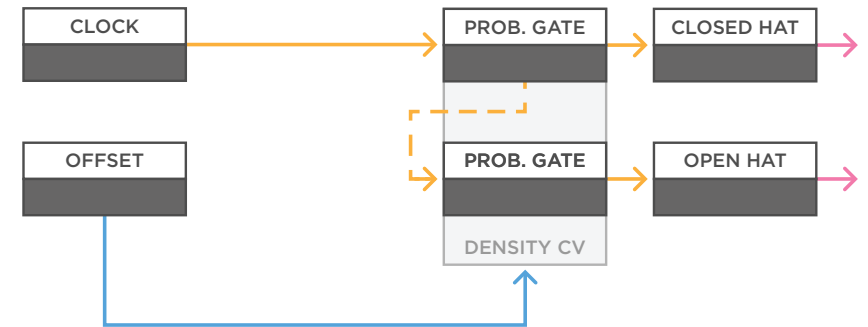
- Audio
- Control Voltage
- Trigger / Gate



(a)

a - The CuteLab MissedOps uses internal connections between the gates. Let's start with the internal normalization in Match mode. Create a trigger pattern, feed the signal directly to the kick, and a copy to the first gate. Send the result of the first gate to a sample that fits nicely on top of the kick. Because of the internal connections, That output is passed on to the second gate as well, and you can use that on something else, for example to trigger another percussive sound that fits nicely on top. By tweaking the density CV with an offset, you can create subtle variations to a steady sound.

MONOTRAIL TECH TALK



(b)

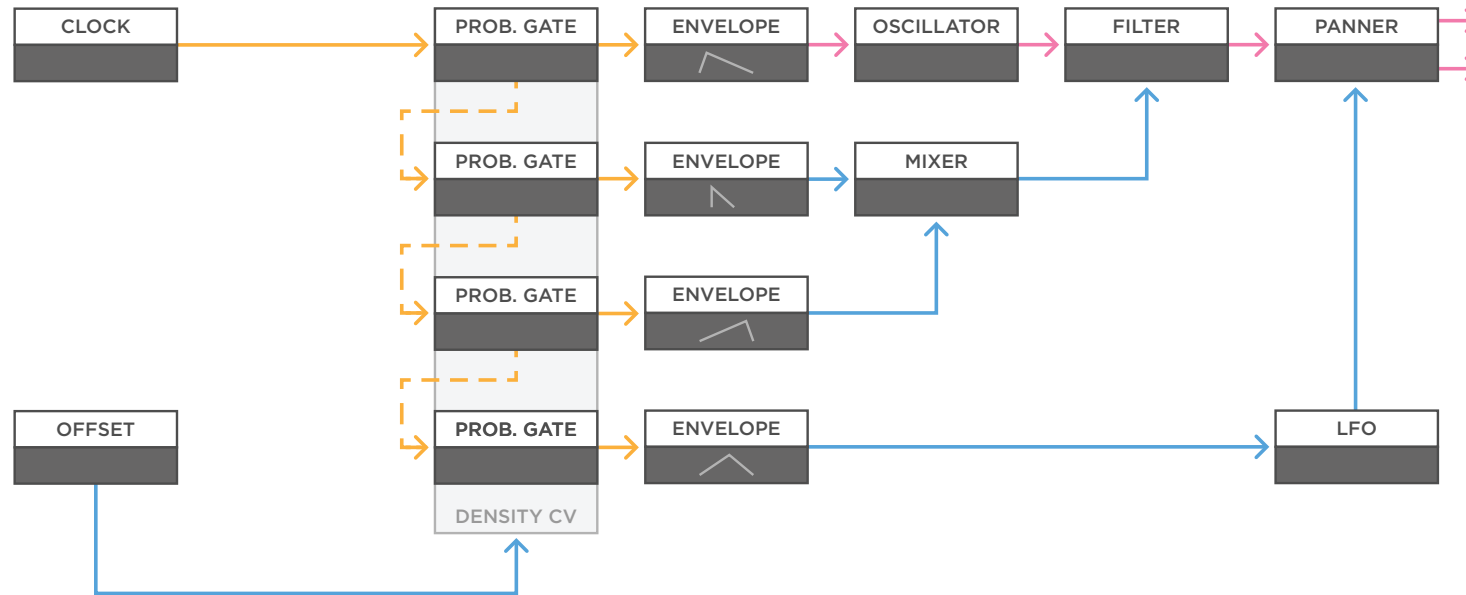
b - The Miss mode can create other interesting setups. For example if you feed a steady clock through a gate, to a closed high hat. Now, every clock pulse blocked by the first gate, is send to the second, so you can use that to trigger an open high hat for example.

Probabilistic Gates - CuteLab MissedOps

Rhythmic Modulation

MONOTRAIL TECH TALK

Audio
Control Voltage
Trigger / Gate



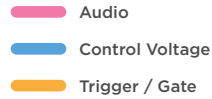
(a)

a - Similar tricks can be used in ambient, generative or drone setups. Especially the Miss setting works nice for this. If you feed a steady clock to the first gate, and send the four outputs to various elements in your patch you can create interesting rhythmic modulation.

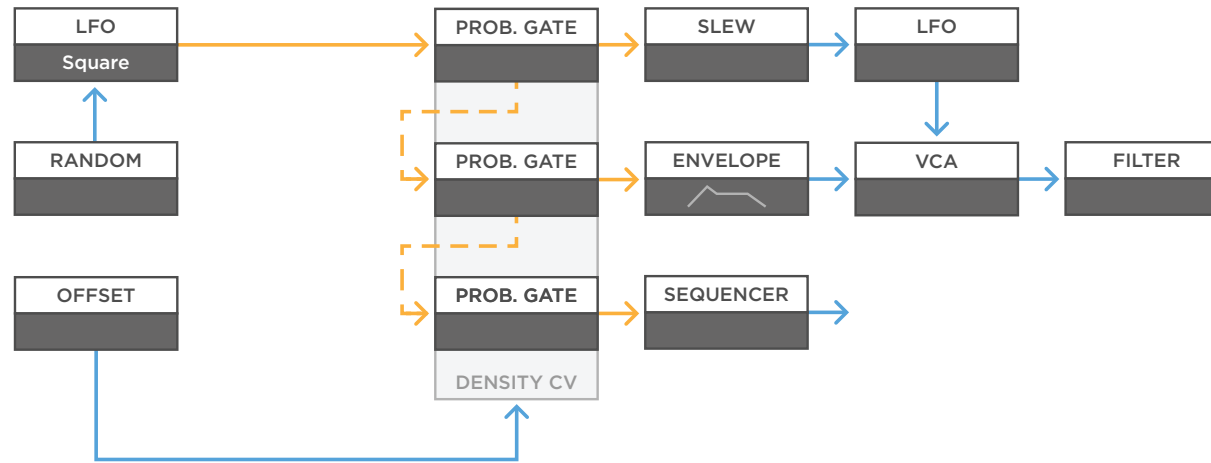
For example, trigger a slow attack decay envelope influencing the waveshape of an oscillator. You can trigger two different envelopes, mix them together and have them influence a filter. Or use an envelope to modulate the speed of an LFO modulating a panning module.

Probabilistic Gates - CuteLab MissedOps

Gate Patch



MONOTRAIL TECH TALK



(a)

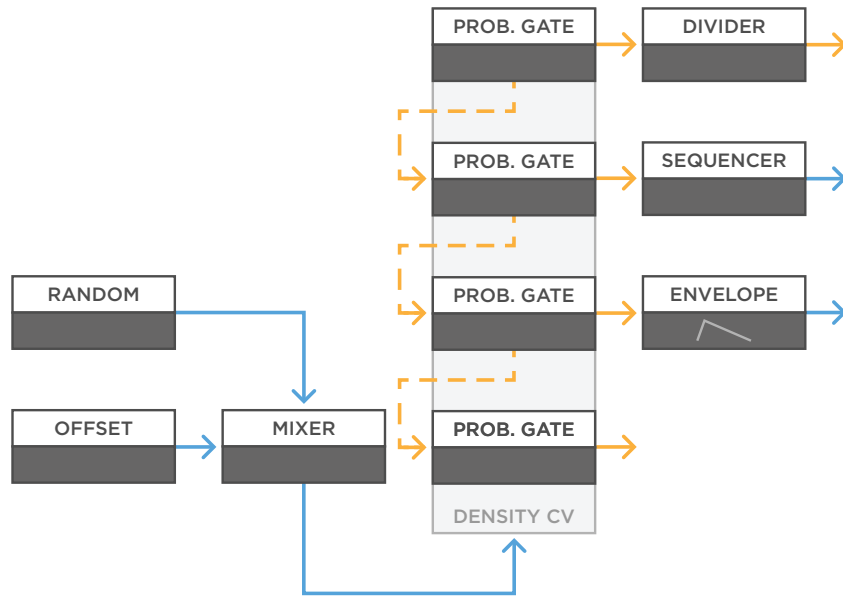
a - Because the module tracks gate length, it can be used to hold values for a longer time. If you feed the module a square wave LFO, and use a slow random voltage to modulate the frequency of that LFO, you get interesting fluctuating gate patterns. You can now use the gate outputs on other elements in your patch. For example, send a gate to an ADSR or slew limiter to change the frequency of an LFO over time.

When you send a gate, slewed gate, or ADSR to a VCA, you can control anything you like. For example, the volume of a sound, or the amount of modulation going from one module to another. You can also use triggers or gates to reset, start or stop a sequencer. And so on.

Probabilistic Gates - CuteLab MissedOps

Random and Reset

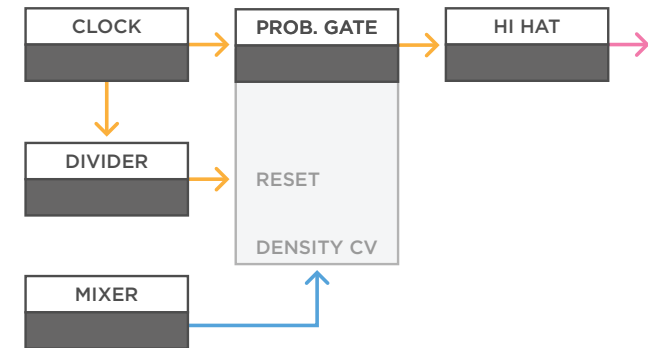
Audio
Control Voltage
Trigger / Gate



(a)

a - The Missed Opportunities also has a built-in trigger generator, that can be used on all sorts of things within a modular system. Only the first channel generates random pulses, but because of the normalization, you still get four different patterns. You can use them as a random clock, to start and stop sequences, or to trigger envelopes and create subtle modulation in an ambient patch for example. Sending a slow random voltage to the density input, and manually switching between the missed and match setting can create interesting variations.

MONOTRAIL TECH TALK



(b)

b - The reset input is fun to experiment with if you like to create semi-random patterns. If you use a steady clock as the input for a gate, and a division of that clock to reset the module, you will get a unique looping pattern. If you create a pattern to trigger some hats, and use something like a 7th division to the reset, in an 8 step beat, you get a looping, but slowly shifting pattern for example.